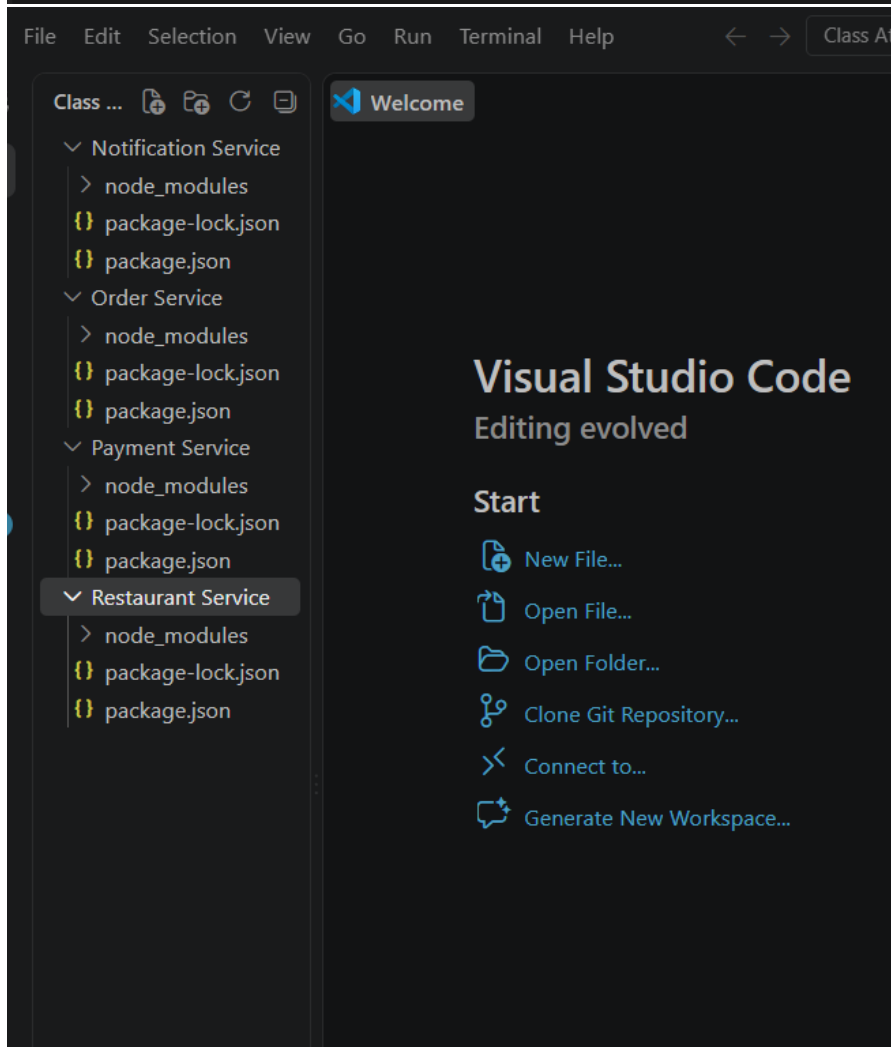
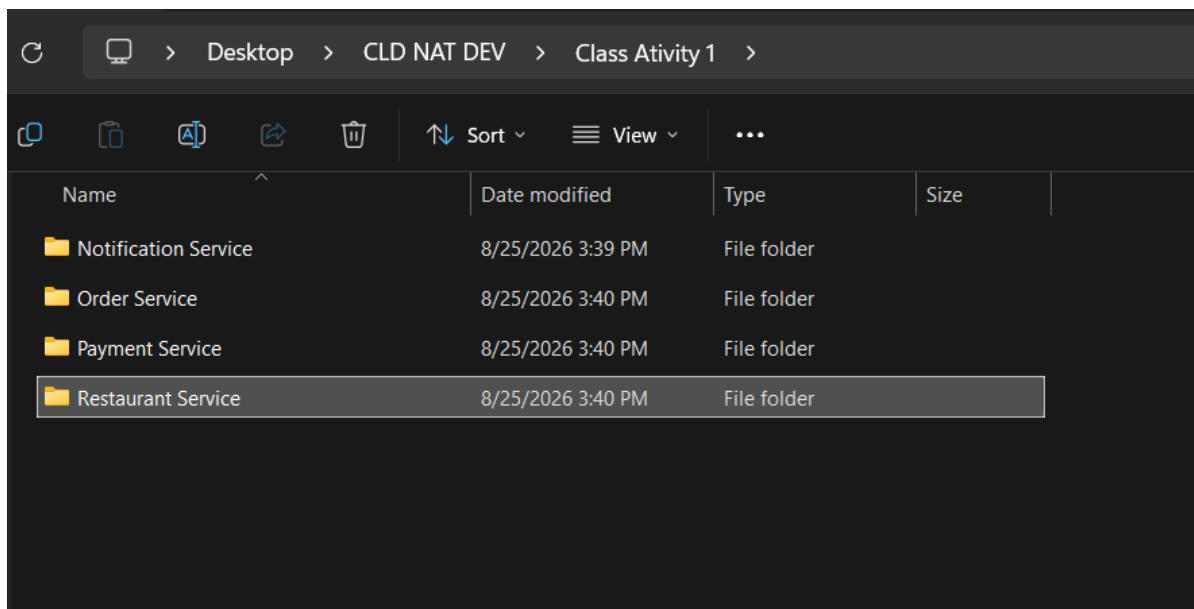




This screenshot shows the VS Code editor with the 'Restaurant Service' tab selected. The file explorer on the left shows a project structure with four services: Notification Service, Order Service, Payment Service, and Restaurant Service. Each service has a 'node_modules' directory and 'index.js', 'package-lock.json', and 'package.json' files. The 'Restaurant Service' directory is expanded, showing these files. The 'index.js' file is open in the editor, displaying the following code:

```
1  const express = require('express');
2  const app = express();
3  app.use(express.json());
4
5  const PORT = 3001;
6
7  // API 1: View all restaurants
8  app.get('/viewallrestaurant', (req, res) => {
9    console.log("viewallrestaurant API called");
10   res.send("viewallrestaurant API called");
11 });
12
13 // API 2: Search restaurant
14 app.get('/searchrestaurant', (req, res) => {
15   console.log("searchrestaurant API called");
16   res.send("searchrestaurant API called");
17 });
18
19 app.listen(PORT, () => {
20   console.log(`Restaurant Service running on http://localhost:${PORT}`);
21 });
22
```

This screenshot shows the VS Code editor with the 'Payment Service' tab selected. The file explorer on the left shows the same project structure as the previous screenshot. The 'Payment Service' directory is expanded, showing 'node_modules', 'index.js', 'package-lock.json', and 'package.json' files. The 'index.js' file is open in the editor, displaying the following code:

```
1  const express = require('express');
2  const app = express();
3  app.use(express.json());
4
5  const PORT = 3003;
6
7  // API: Process payment
8  app.post('/paymentprocess', (req, res) => {
9    console.log("paymentprocess API called");
10
11    // Simple Success/Failure simulation (kept minimal, needed for the Order flow)
12    const status = Math.random() > 0.5 ? "Success" : "Failure";
13    res.json({ status: status });
14 });
15
16 app.listen(PORT, () => {
17   console.log(`Payment Service running on http://localhost:${PORT}`);
18 });
19
```

This screenshot shows the VS Code editor with the 'Notification Service' tab selected. The file explorer on the left shows the same project structure. The 'Notification Service' directory is expanded, showing 'node_modules', 'index.js', 'package-lock.json', and 'package.json' files. The 'index.js' file is open in the editor, displaying the following code:

```
1  const express = require('express');
2  const app = express();
3  app.use(express.json());
4
5  const PORT = 3004;
6
7  // API: Send notification
8  app.post('/sendnotification', (req, res) => {
9    console.log("sendnotification API called");
10
11    const { status } = req.body;
12    res.send(`sendnotification API called - Order status: ${status}`);
13 });
14
15 app.listen(PORT, () => {
16   console.log(`Notification Service running on http://localhost:${PORT}`);
17 });
18
```

```
File Edit Selection View Go Run Terminal Help ← → Class Activity 1

Class ... JS index.js Restaurant Service JS index.js Order Service JS index.js Payment Service JS index.js Notification Service

Notification Service
  > node_modules
  JS index.js
  {} package-lock.json
  {} package.json
Order Service
  > node_modules
  JS index.js
  {} package-lock.json
  {} package.json
Payment Service
  > node_modules
  JS index.js
  {} package-lock.json
  {} package.json
Restaurant Service
  > node_modules
  JS index.js
  {} package-lock.json
  {} package.json
Class Activity 1 - Mi...

Order Service > JS index.js > ...
1  const express = require('express');
2  const axios = require('axios');
3  const app = express();
4  app.use(express.json());
5
6  const PORT = 3002;
7  const PAYMENT_SERVICE_URL = "http://localhost:3003/paymentprocess";
8  const NOTIFICATION_SERVICE_URL = "http://localhost:3004/sendnotification";
9
10 // API 1: Add order
11 // Flow: Order API -> Payment API -> Order API -> Notification API -> User
12 app.post('/addorder', async (req, res) => {
13   console.log("addorder API called");
14
15   try {
16     // Step 1: Call Payment Process API
17     console.log("Order API calling Payment API...");
18     const paymentResponse = await axios.post(PAYMENT_SERVICE_URL, req.body);
19     const paymentStatus = paymentResponse.data.status; // "Success" or "Failure"
20     console.log("Payment API responded with: ${paymentStatus}");
21
22     // Step 2: Call Notification Service API with the payment result
23     console.log("Order API calling Notification API...");
24     const notificationResponse = await axios.post(NOTIFICATION_SERVICE_URL, { status: paymentStatus });
25     console.log("Notification API responded:", notificationResponse.data);
26
27     // Step 3: Return final result to the User
28     res.send("addorder API called - Payment: ${paymentStatus}");
29
30   } catch (error) {
31     console.error("Error during order processing:", error.message);
32     res.status(500).send("Error processing order");
33   }
34 });
35
36 // API 2: View order
37 app.get('/vieworder', (req, res) => {
38   console.log("vieworder API called");
39   res.send("vieworder API called");
40 });
41
42 // API 3: Cancel order
43 app.post('/cancelorder', (req, res) => {
44   console.log("cancelorder API called");
45   res.send("cancelorder API called");
46 });
47
48 app.listen(PORT, () => {
49   console.log("Order Service running on http://localhost:${PORT}");
50 });
```

```
Windows PowerShell
PS C:\Users\User\OneDrive - American University of Phnom Penh\Desktop\CLD NAT DEV\Class Activity 1\Restaurant Service> node index.js
Restaurant Service running on http://localhost:3001
```

```
Windows PowerShell
PS C:\Users\User\OneDrive - American University of Phnom Penh\Desktop\CLD NAT DEV\Class Activity 1\Order Service> node index.js
Order Service running on http://localhost:3002
```

```
Windows PowerShell
PS C:\Users\User\OneDrive - American University of Phnom Penh\Desktop\CLD NAT DEV\Class Activity 1\Payment Service> node index.js
Payment Service running on http://localhost:3003
```

```
Windows PowerShell
PS C:\Users\User\OneDrive - American University of Phnom Penh\Desktop\CLD NAT DEV\Class Activity 1\Notification Service>n
ode index.js
Notification Service running on http://localhost:3004
```

My Collection > Get data

GET http://localhost:3001/viewallrestaurant

Send

Docs Params Authorization Headers 6 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body Cookies Headers 7 Test Results 1/1

200 OK • 4 ms • 256 B • Save Response

HTML Preview Visualize

1 viewallrestaurant API called

Search Invite Upgrade

Overview Overview Getting started GET http://localhost:50 GET Get data No environment

My Collection > Get data

GET http://localhost:3001/searchrestaurant

Send

Docs Params Authorization Headers 6 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body Cookies Headers 7 Test Results 1/1

200 OK • 5 ms • 255 B • Save Response

HTML Preview Visualize

1 searchrestaurant API called

Overview Overview Getting started GET http://localhost:50 GET Get data No environment

My Collection > Get data

GET http://localhost:3002/vieworder

Send

Docs Params Authorization Headers 6 Body Scripts Tests Settings Cookie

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body Cookies Headers 7 Test Results 1/1

200 OK • 15 ms • 248 B • Save Response

HTML Preview Visualize

1 vieworder API called

Overview Overview Getting started GET http://localhost:50(POST Get data + No environment

My Collection > Get data Save Share

POST http://localhost:3002/cancelorder Send

Docs Params Authorization Headers 7 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies Headers 7 Test Results 1/1 200 OK 8 ms 250 B Save Response

HTML Preview Visualize

1 cancelorder API called

Overview Overview Getting started GET http://localhost:50(POST Get data + No environment

My Collection > Get data Save Share

POST http://localhost:3002/addorder Send

Docs Params Authorization Headers 7 Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

Body Cookies Headers 7 Test Results 1/1 200 OK 600 ms 266 B Save Response

HTML Preview Visualize

1 addorder API called - Payment: Success

My Collection > Get data Save Share

POST http://localhost:3002/addorder Send

Docs Params Authorization Headers 8 Body Scripts Tests Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Schema Beautify

```
1 {
2   "orderId": 1,
3   "amount": 15
4 }
```

Body Cookies Headers 7 Test Results 1/1 200 OK 204 ms 266 B Save Response

HTML Preview Visualize

1 addorder API called - Payment: Success

My Collection > Get data

POST http://localhost:3003/paymentprocess

Docs Params Authorization Headers 8 Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "orderId": 1,
3   "amount": 15
4 }
```

Body Cookies Headers 7 Test Results 1/1 200 OK • 10 ms • 255 B Save Response

{ JSON Preview Visualize

```
1 {
2   "status": "Failure"
3 }
```

Overview Overview Getting started GET http://localhost:50 POST Get data No environment

My Collection > Get data

POST http://localhost:3004/sendnotification

Docs Params Authorization Headers 8 Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "status": "Success"
3 }
```

Body Cookies Headers 7 Test Results 1/1 200 OK • 9 ms • 279 B Save Response

HTML Preview Visualize

```
1 sendnotification API called - Order status: Success
```

Windows PowerShell

```
PS C:\Users\User\OneDrive - American University of Phnom Penh\Desktop\CLD NAT DEV\Class Activity 1\Order Service> node in
dex.js
Order Service running on http://localhost:3002
vieworder API called
vieworder API called
addorder API called
Order API calling Payment API...
Payment API responded with: Success
Order API calling Notification API...
Notification API responded: sendnotification API called - Order status: Success
cancelorder API called
cancelorder API called
cancelorder API called
addorder API called
Order API calling Payment API...
Payment API responded with: Success
Order API calling Notification API...
Notification API responded: sendnotification API called - Order status: Success
```

File Edit Selection View Go Run Terminal Help

Class Activity 1

Class ... JS index.js Restaurant Service JS index.js Order Service JS index.js Payment Service JS index.js Notification Service

Notification Service

node_modules

JS index.js

package-lock.json

package.json

Order Service

node_modules

JS index.js

package-lock.json

package.json

Payment Service

node_modules

JS index.js

package-lock.json

package.json

Restaurant Service

node_modules

JS index.js

package-lock.json

package.json

Class Activity 1 - ML...

Order Service > JS index.js > ...

```
1 const express = require('express');
2 const axios = require('axios');
3 const app = express();
4 app.use(express.json());
5
6 const PORT = 3002;
7 const PAYMENT_SERVICE_URL = "http://localhost:3003/paymentprocess";
8 const NOTIFICATION_SERVICE_URL = "http://localhost:3004/sendnotification";
9
10 // API 1: Add order
11 // Flow: Order API -> Payment API -> Order API -> Notification API -> User
12 app.post('/addorder', async (req, res) => {
13   console.log("addorder API called");
14
15   try {
16     // Step 1: Call Payment Process API
17     console.log("Order API calling Payment API...");
18     const paymentResponse = await axios.post(PAYMENT_SERVICE_URL, req.body);
19     const paymentStatus = paymentResponse.data.status; // "Success" or "Failure"
20     console.log("Payment API responded with: ${paymentStatus}");
21
22     // Step 2: Call Notification Service API with the payment result
23     console.log("Order API calling Notification API...");
24     const notificationResponse = await axios.post(NOTIFICATION_SERVICE_URL, { status: paymentStatus });
25     console.log("Notification API responded:", notificationResponse.data);
26
27     // Step 3: Return full result to the User
28     res.json({
29       message: "addorder API called",
30       order: req.body,
31       paymentResult: paymentResponse.data,
32       notificationResult: notificationResponse.data
33     });
34
35   } catch (error) {
36     console.error("Error during order processing:", error.message);
37     res.status(500).send("Error processing order");
38   }
39 });
40
41 // API 2: View order
42 app.get('/vieworder', (req, res) => {
43   console.log("vieworder API called");
44   res.send("vieworder API called");
45 });
46
47 // API 3: Cancel order
48 app.post('/cancelorder', (req, res) => {
49   console.log("cancelorder API called");
50   res.send("cancelorder API called");
51 });
52
53 app.listen(PORT, () => {
54   console.log('Order Service running on http://localhost:${PORT}');
55 });
56
```

Overview Overview Getting started GET http://localhost:5000/a POST Get data No environment

My Collection > Get data Save Share

POST http://localhost:3002/addorder Send

Docs Params Authorization Headers 8 Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "orderId": 1,
3   "amount": 15
4 }
```

Body Cookies Headers 7 Test Results 1/1 200 OK - 140 ms - 415 B Save Response

JSON Preview Visualize

```
1 {
2   "message": "addorder API called",
3   "order": {
4     "orderId": 1,
5     "amount": 15
6   },
7   "paymentResult": {
8     "status": "Success"
9   },
10  "notificationResult": "sendnotification API called - Order status: Success"
11 }
```